

Suyeon Choi

Curriculum Vitae

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Appointments

2/2026- **Assistant Professor**, *Computer Science and Engineering*, Seoul National University, Korea.

Education

- 9/2020–9/2024 **Ph.D.**, *Electrical Engineering*, Stanford University, Stanford, CA.
Thesis title: Design of holographic display systems based on artificial intelligence
Committee: Gordon Wetzstein, David Miller, Olav Solgaard, Kayvon Fatahalian, Mark Horowitz, Douglas Lanman
- 9/2019–6/2021 **M.Sc.**, *Electrical Engineering*, Stanford University, Stanford, CA.
Advisor: Gordon Wetzstein
- 3/2013–2/2019 **B.Sc.**, *Electrical Engineering*, Seoul National University, Seoul, Korea.
Advisors: Deog-kyoon Jeong, Soo-Mook Moon, and Byoungho Lee.

Professional Experience

- 10/2024–1/2026 **Postdoctoral Scholar**, *Electrical Engineering*, Stanford University, Stanford, CA.
Advisor: Gordon Wetzstein
- 7/2022–9/2022 **Intern**, *Display Systems Research Team (DSR)*, Meta Reality Labs, Redmond, WA.
Worked with Changwon Jang, Andrew Maimone, and Douglas Lanman.
- 7/2020–9/2020 **Intern**, *New Experience Team (NXP)*, NVIDIA, Santa Clara, CA.
Worked with Jonghyun Kim, Ward Lopes and David Luebke.

Honors and Awards

- 2026 Creative-Pioneering Researchers Program, SNU, Korea
- 2026 Outstanding Young Scientist Grants, NRF, Korea
- 2025 ACM SIGGRAPH Outstanding Doctoral Dissertation Award Honorable Mention
- 2023 SPIE Optics and Photonics Education Scholarship
- 2023 Frontiers of Science Award, International Congress of Basic Science [4]
- 2022 Meta Research PhD Fellowship - AR/VR Computer Graphics
- 2022 NVIDIA Graduate Fellowship Finalist
- 2020 School of Engineering Fellowship, Stanford University
- 2019 Kwanjeong Scholarship
- 2019 Korea Government Scholarship
- 2019 Yongwoon Scholarship, declined due to period overlap with other PhD scholarships.
- 2019 2nd Prize, SPIE Student Optical Design Challenge 2019 [C1]
- 2013 Presidential Science Scholarship, South Korea
- 2012 Silver Medal, the 43rd International Physics Olympiad (IPhO)

Publications

*denotes equal contribution.

Peer-reviewed Journals and Technical Papers in Conferences

- [16] **S. Choi***, C. Jang*, D. Lanman, and G. Wetzstein, "Synthetic aperture waveguide holography for compact mixed reality displays with large étendue", *Nature Photonics*, 2025.
- [15] **S. Choi***, B. Chao*, J. Yang, M. Gopakumar, and G. Wetzstein, "Gaussian wave splatting for computer-generated holography", *ACM Transactions on Graphics (Proc. SIGGRAPH 2025)*.
- [14] B. Chao, M. Gopakumar, **S. Choi**, J. Kim, L. Shi, and G. Wetzstein, "Large Étendue 3D Holographic Display with Content-Adaptive Dynamic Fourier Modulation", *Proc. SIGGRAPH Asia 2024*.
- [13] M. Gopakumar*, G. Y. Lee*, **S. Choi**, B. Chao, Y. Peng, J. Kim, G. Wetzstein, "Full-color 3D holographic augmented reality displays with metasurface waveguides", *Nature*, 2024
- [12] D. Kim*, S. W. Nam*, **S. Choi***, J. M. Seo, G. Wetzstein, and Y. Jeong, "Holographic parallax improves 3D perceptual realism", *ACM Transactions on Graphics (Proc. SIGGRAPH 2024)*.
- [11] B. Chao, M. Gopakumar, **S. Choi**, and G. Wetzstein, "High-Brightness Holographic Projection", *Optics Letters*, 2023.
- [10] **S. Choi***, M. Gopakumar*, Y. Peng, J. Kim, M. O'Toole, and G. Wetzstein, "Time-multiplexed Neural Holography: A Flexible Framework for Holographic Near-eye Displays with Fast Heavily-quantized Spatial Light Modulators", *Proc. SIGGRAPH 2022*.
- [9] J. Kim, M. Gopakumar, **S. Choi**, Y. Peng, W. Lopes, and G. Wetzstein, "Holographic glasses for Virtual Reality", *Proc. SIGGRAPH 2022*.
- [8] **S. Choi***, M. Gopakumar*, Y. Peng, J. Kim, and G. Wetzstein, "Neural 3D Holography: Learning Accurate Wave Propagation Models for 3D Holographic Virtual and Augmented Reality Displays", *ACM Transactions on Graphics (Proc. SIGGRAPH Asia 2021)*.
- [7] Y. Peng*, **S. Choi***, J. Kim, and G. Wetzstein, "Speckle-free Holography with Partially Coherent Light Sources and Camera-in-the-loop Calibration", *Science Advances*, 2021.
- [6] M. Gopakumar, J. Kim, **S. Choi**, Y. Peng, and G. Wetzstein, "Unfiltered Holography: Optimizing High Diffraction Orders without Optical Filtering for Compact Holographic Displays", *Optics Letters*, 2021
- [5] **S. Choi**, J. Kim, Y. Peng, and G. Wetzstein "Optimizing image quality for holographic near-eye displays with Michelson Holography", *Optica*, 2021.
- [4] Y. Peng, **S. Choi**, N. Padmanaban, and G. Wetzstein "Neural Holography with Camera-in-the-loop Training", *ACM Transactions on Graphics (Proc. SIGGRAPH Asia 2020)*.
- [3] D. Yoo*, S. Lee*, Y. Jo, J. Cho, **S. Choi**, and B. Lee "Volumetric Head-Mounted Display with Locally Adaptive Focal Blocks", *IEEE Transactions on Visualization and Computer Graphics*, 2022.
- [2] Y. Jo*, S. Lee*, D. Yoo, **S. Choi**, D. Kim, and B. Lee, "Tomographic Projector: Large Scale Volumetric Display with Uniform Viewing Experiences", *ACM Transactions on Graphics (Proc. SIGGRAPH Asia 2019)*.
- [1] **S. Choi**, S. Lee, Y. Jo, D. Yoo, D. Kim, and B. Lee, "Optimal Binary Representation via Non-convex Optimization on Tomographic Displays", *Optics Express*, 2019.

Conference Proceedings - Others

- [C2] **S. Choi**, Y. Peng, J. Kim, and G. Wetzstein "High-quality holographic displays using double SLMs and camera-in-the-loop optimization", Proc. SPIE 11765, *Optical Architectures for Displays and Sensing in Augmented, Virtual, and Mixed Reality (AR, VR, MR) II*, 2021.
- [C1] D. Yoo*, S. Lee*, Y. Jo, J. Cho, **S. Choi**, and B. Lee, "15 focal planes head-mounted display using LED array backlight", Proc. SPIE 11040, *SPIE Photonics West Student Optical Design Challenge*, 2019.

Conference Proceedings - Invited papers

- [17] **S. Choi**, B. Chao, G.-Y. Lee, C. Jang, D. Lanman, G. Wetzstein, "Holographic rendering algorithms and display architectures with large étendue", Proc. SPIE 13917, Practical Holography XXXX: Displays, Materials, and Applications, (SPIE Photonics West 2026). **(Invited Paper)**
- [16] **S. Choi**, B. Chao, J. Yang, S.-W. Nam, G. Wetzstein, "Artificial Intelligence-based Holographic Glasses", in *SID Symposium Digest of Technical Papers, 2026* **(Invited Paper)**
- [15] G.-Y. Lee, M. Gopakumar, **S. Choi**, B. Chao, Y. Peng, J. Kim, G. Wetzstein, "Nanophotonics and AI for Augmented Reality and Imaging Applications", in *SID Symposium Digest of Technical Papers, 2025* **(Invited Paper)**
- [14] **S. Choi**, M. Gopakumar, Y. Peng, J. Kim, M. O'Toole, and G. Wetzstein, "Partially coherent neural holography with fast spatial light modulators," Proc. SPIE 12435, Emerging Digital Micromirror Device Based Systems and Applications XV (SPIE Photonics West 2023). **(Invited Paper)**
- [13] M. Gopakumar, **S. Choi**, J. Kim, Y. Peng, and G. Wetzstein, "Enabling ultra-compact, high-quality 3D displays with neural holography", Proc. SPIE 12445, Practical Holography XXXVII: Displays, Materials, and Applications, (SPIE Photonics West 2023). **(Invited Paper)**
- [12] **S. Choi**, Y. Peng, M. Gopakumar, J. Kim, G. Wetzstein, "Enabling Augmented-Reality Near-Eye and Head-Up Displays with Neural Holography", in *SID Symposium Digest of Technical Papers, 2022* **(Invited Paper)**
- [11] M. Gopakumar, Y. Peng, **S. Choi**, J. Kim, G. Wetzstein, "Advances in Neural Holographic Displays for Virtual and Augmented Reality", in *SID Symposium Digest of Technical Papers, 2022* **(Invited Paper)**

Public Demonstrations

- 2024 **Holographic Parallax**, S.-W. Nam*, D. Kim*, S. Choi*, M. Gopakumar, J. Lee, S. Lee, B. Chao, G. Wetzstein, and Y. Jeong, ACM SIGGRAPH 2024 Emerging Technologies.
- 2023 **Neural Holographic Near-eye Displays for Virtual Reality**, S. Choi*, M. Gopakumar*, B. Chao*, G. Y. Lee, J. Kim, G. Wetzstein, ACM SIGGRAPH 2023 Emerging Technologies.
- 2020 **Neural Holography**, Y. Peng, S. Choi, N. Padmanaban, J. Kim, G. Wetzstein, ACM SIGGRAPH 2020 Emerging Technologies.

Keynotes and Invited Talks

- 08/2026 Design of Holographic Display Systems based on Artificial Intelligence, *International Meeting on Information Display*, Busan, Korea.
- 08/2026 Design of Holographic Display Systems based on Artificial Intelligence, *Plenary talk, 2026 International Congress of Basic Science (ICBS)*, Beijing, China.
- 05/2026 Tutorial for AI-based holographic displays, *The 33rd Conference on Optoelectronics and Optical Communications*, Busan, Korea.
- 05/2026 AI-based holographic rendering algorithms and display architectures, *SID Display Week*, Los Angeles, CA.
- 04/2026 Design of Holographic Display Systems based on Artificial Intelligence, *Asiagraphics Workshop on Intelligent Graphics*, Seoul, Korea.
- 03/2026 Keynote, AI-based holographic glasses for AR and VR, *International Workshop on Virtual Reality for Human and Spatial Augmentation (VR-HSA)*, in conjunction with IEEE VR 2026, Daegu, Korea
- 03/2026 Synthetic Aperture Waveguide Holography and Gaussian Wave Splatting, *Daegu Gyeongbuk Institute of Science and Technology (DGIST)*, Daegu, Korea.
- 03/2026 Synthetic Aperture Waveguide Holography and Gaussian Wave Splatting, *Electronics and Telecommunications Research Institute (ETRI)*, Daejeon, Korea.

- 01/2026 Synthetic Aperture Waveguide Holography and Gaussian Wave Splatting, *SPIE Practical Holography XXXX: Displays, Materials, and Applications*, San Francisco, CA.
- 01/2026 Synthetic Aperture Waveguide Holography and Gaussian Wave Splatting, *SIGGRAPH Silicon Valley Chapter*, Santa Clara, CA.
- 12/2025 AI-based holographic glasses, *SNU AI Computing School*, Seoul, Korea
- 11/2025 Holographic displays based on artificial intelligence, *SCIEN Seminar Series*, Stanford, CA.
- 10/2025 Design of Holographic Display Systems based on Artificial Intelligence, *Optica Frontiers in Optics + Laser Sciencs (FiO+LS)*, Denver, CO.
- 08/2025 Design of Holographic Display Systems based on Artificial Intelligence, *Graphics and Mixed Environment Seminar (GAMES)*, Virtual.
- 08/2025 Design of Holographic Display Systems based on Artificial Intelligence, *Samsung Research*, Seoul, Korea (Virtual).
- 07/2025 Design of Holographic Display Systems based on Artificial Intelligence, *Korea Electronics Technology Institute (KETI)*, Seoul, Korea (Virtual).
- 06/2025 Design of Holographic Display Systems based on Artificial Intelligence, *CVPR Computational Cameras and Displays Workshop*, Nashville, TN.
- 06/2025 Design of Holographic Display Systems based on Artificial Intelligence, *SPIE Digital Optical Technologies (DOT) Conference*, Munich, Germany.
- 03/2025 Design of Holographic Display Systems based on Artificial Intelligence, *SID Bay Area Chapter / SCIEN Colloquium and EE 292E*, Sunnyvale, CA.
- 09/2024 Design of Holographic Display Systems based on Artificial Intelligence, *SIGGRAPH Silicon Valley Chapter*, Santa Clara, CA.
- 07/2024 Design of Holographic Display Systems based on Artificial Intelligence, *Hyundai NGV*, Virtual.
- 10/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *Optica Incubator Meetings (Optical Waveguides: A Key to Socially Acceptable AR Glasses?)*, Tacoma, WA.
- 09/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *University College London, High-beams Seminar*, Virtual.
- 09/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *Applied Materials*, Santa Clara, CA.
- 02/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *Seoul National University (SNU), Dept. of Computer Science*, Seoul, Korea.
- 02/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *KAIST, Dept. of Computer Science*, Daejeon, Korea.
- 02/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *KAIST, Dept. of Physics*, Daejeon, Korea.
- 02/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *Pohang University of Science and Technology (POSTECH)*, Pohang, Korea.
- 02/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *Electronics and Telecommunications Research Institute (ETRI)*, Daejeon, Korea.
- 02/2023 Partially-Coherent Neural Holography and Holographic Glasses, *Annual Conference of Optical Society of Korea*, Busan, Korea.
- 02/2023 Partially-Coherent Neural Holography, *SPIE Photonics West, Emerging Digital Micromirror Device Based Systems and Applications XV*, San Francisco, CA.
- 11/2022 Neural Holography for Next-generation AR/VR Display Systems, *2022 SystemX Fall Conference*, Stanford, CA.
- 09/2022 Partially-Coherent Neural Holography and Holographic Glasses, *Optica Virtual panel discussion, "Could Deep Learning Improve Visual Quality in Holographic Displays?"*, Virtual.

- 07/2022 Partially-Coherent Neural Holography and Holographic Glasses, *Optica 3D Image Acquisition and Display: Technology, Perception and Applications: Deep Learning and Machine Learning for 3D Imaging*, Vancouver, Canada.
- 05/2022 Enabling Augmented-Reality Near-Eye and Head-Up Displays with Neural Holography, *SID Display Week*, San Jose, CA.
- 12/2021 Neural Holography Pro: Computationally Enabling Compact, High-quality 3D Holographic Displays, *Graphics and Mixed Environment Seminar (GAMES)*, Virtual.
- 11/2021 Enabling Next-generation Holographic Displays with Artificial Intelligence, *Optica Frontiers in Optics LS AR/VR*, Virtual.
- 02/2021 High-quality holographic displays using double SLMs, *SPIE AR,VR,MR Technical Talks*, Virtual.
- 12/2020 Neural Holography: High-quality, Real-time Computer-generated Holographic Displays, *Graphics and Mixed Environment Seminar (GAMES)*, Virtual.

Media Coverage

- 2025 **Goodbye, Bulky VR. Hello, Holographic Glasses**, Tech Explorist
- 2025 **A leap toward lighter, sleeker mixed reality displays**, Stanford News
- 2024 **AI and holographic imaging bring 3D augmented reality to regular glasses**, Stanford News
- 2022 **Nvidia, Stanford University propose thin and light holographic glasses**, ZDNET
- 2021 **Stanford researchers are using artificial intelligence to create better virtual reality experiences**, Stanford News
- 2021 **Improvements to Holographic Displays Poised to Enhance Virtual and Augmented Reality**, Optica Newsroom
- 2020 **Using AI to Revolutionize Real-Time Holography**, Stanford HAI News

Teaching Experience

Instructor

- Fall 2026 Introduction to Machine Learning, *SNU*
- Spring 2026 Topics in Computer Graphics: Computational Imaging, *SNU*

Guest Lectures

- 05/2025 Design of Holographic Display Systems based on Artificial Intelligence, *Stanford University, CEE 342: Designing for Gradient Spaces, Stanford, CA*
- 05/2025 Design of Holographic Display Systems based on Artificial Intelligence, *Stanford University, EE 267: Virtual Reality, Stanford, CA*
- 11/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *University of Utah, ECE 5331 - Optics for Energy Course Guest Seminar, UT*
- 10/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *the College of New Jersey (TCNJ), the CS Department Colloquia Series, NJ*
- 10/2023 Neural Holography for Next-generation Virtual and Augmented Reality Displays, *Sungkyunkwan University (SKKU), Dept. of Immersive Media Engineering Colloquia Series, Seoul.*

Teaching Assistant

- Spring 23-24, Spring 21-22, Spring 2020-21 EE267: Virtual Reality, Stanford University

Undergraduate Teaching Assistant

- Fall 2018 Digital Systems Design and Experiments, Seoul National University
- Fall 2017 Introduction to Electromagnetism, Seoul National University
- Spring 2015 Introduction to Circuit Theory and Laboratory, Seoul National University
- Fall 2014 Digital Logic Design and Lab, Seoul National University

Tutor

Spring/Fall 2018, Basic Physics, Seoul National University
Spr 2015, Fall 2014

2013-2015 Math, Seoul National University Children's Hospital

Services

Program Committee ACM SIGGRAPH, Technical Papers, 2026, 2027
ACM SIGGRAPH Asia, Technical Papers, 2025, 2026
IEEE Int. Conference on Virtual Reality and 3D User Interfaces (VR), 2026, 2027
IEEE Int. Conference on Computational Photography (ICCP), 2025 –
IEEE Int. Symposium on Mixed and Augmented Reality (ISMAR), 2023 – 2026
ACM Symposium on Virtual Reality Software and Technology (VRST), 2025
ACM SIGGRAPH, Posters, 2026
ACM SIGGRAPH Asia, XR, 2025
Eurographics, Short Papers, 2026

Reviewer Nature Photonics, Nature Communications, Nature Machine Intelligence, Science Advances,
ACM SIGGRAPH, ACM SIGGRAPH Asia, ACM Transactions on Graphics (TOG), NeurIPS,
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Transactions
on Computational Imaging (TCI), IEEE Transactions on Visualization and Computer Graphics
(TVCG), IEEE VR, IEEE ISMAR, Light: Science & Applications, Optica, Photonics Research,
Optics Letters, Optics Express, Applied Optics, Optics Continuum, Laser & Photonics Reviews,
Virtual Reality, Scientific Reports, Eurographics

Member ACM SIGGRAPH, SPIE, SID, Optica

Student Reader Stanford EE Masters Admissions 2023-2024